Live Gold & Silver Price Prediction using Machine Learning

A Mini Project in Machine Learning

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Introduction

- Gold and silver are the two most tracked precious metals, priced in US dollars per troy ounce.
- Their prices move with demand, inflation, the dollar and global market sentiment.
- This project fetches the LIVE price of both metals and converts it into Indian Rupees per gram.
- A Machine Learning model then forecasts the price for the next 12 months.
- Everything is shown on one web dashboard, so the numbers can be read without any training.

Problem Statement

- Given the historical monthly price of gold and silver, predict their price for the coming months.
- This is a supervised regression problem: past behaviour is the input, future price is the output.
- Several models are trained, scored on data they have never seen, and the best one is kept.
- The winning model is connected to a live price feed and presented on a dashboard in rupees.

Objectives

- Build a clean monthly price dataset for gold and silver from the year 2000 up to today.
- Read the live price of both metals and the US dollar to rupee rate from public APIs.
- Engineer features that capture short-term momentum instead of raw price levels.
- Compare Random Forest, Linear Regression and Ridge against a naive baseline.
- Deliver an attractive dashboard showing present and future prices in Indian Rupees.

Live Data Sources

- api.gold-api.com/price/XAU - live gold price in US dollars per troy ounce.
- api.gold-api.com/price/XAG - live silver price in US dollars per troy ounce.
- open.er-api.com - today's US dollar to Indian rupee exchange rate.
- All three are free public feeds, so no key or login is needed.
- Conversion: Rs per gram = US$ per ounce x USD-INR rate / 31.1035 (one troy ounce = 31.1035 g).

Dataset

- Two files: gold_prices.csv and silver_prices.csv - one row per month since January 2000.
- Columns: Date, Year, Month and Price (US dollars per troy ounce).
- Yearly anchors are approximate LBMA annual averages, interpolated month to month with momentum.
- Model features: the last 3 monthly returns, plus the month and the year.
- The live API price is appended as the latest point, so the forecast always starts from today.

Methodology

- The model predicts the month-to-month RETURN, never the raw price.
- Why: a tree model cannot guess a price higher than anything it saw, so a 25 year rising trend breaks it.
- Price is rebuilt as next price = last price x (1 + predicted return).
- Forecasting rolls forward one month at a time, feeding each answer back as the next input.
- The split is chronological - the last 60 months are held out and never used for fitting.

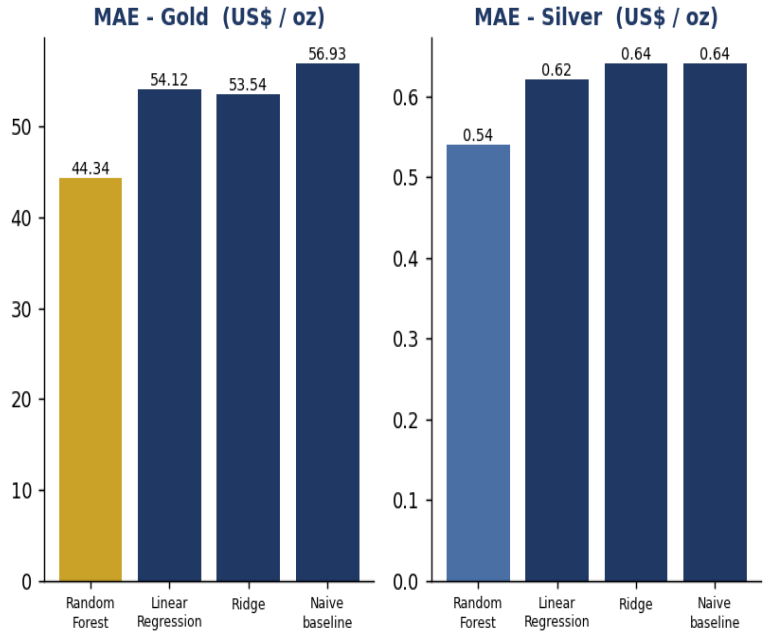
Technology Used

- Language: Python.
- Data handling: pandas and NumPy.
- Models: scikit-learn - Random Forest, Linear Regression, Ridge.
- Backend: Flask, serving one JSON summary to the page.
- Frontend: Chart.js on a dark dashboard. Model save / load: Joblib.

Results

Model	Gold MAE (US$/oz)	Gold R-sq	Silver MAE (US$/oz)	Silver R-sq
Random Forest	44.34	0.9910	0.54	0.9942
Linear Regression	54.12	0.9889	0.62	0.9909
Ridge	53.54	0.9889	0.64	0.9903
Naive baseline	56.93	0.9875	0.64	0.9897

Random Forest wins on both metals and beats the naive baseline. Test window: last 60 months, from Oct 2021. Lower MAE and higher R-squared are better.

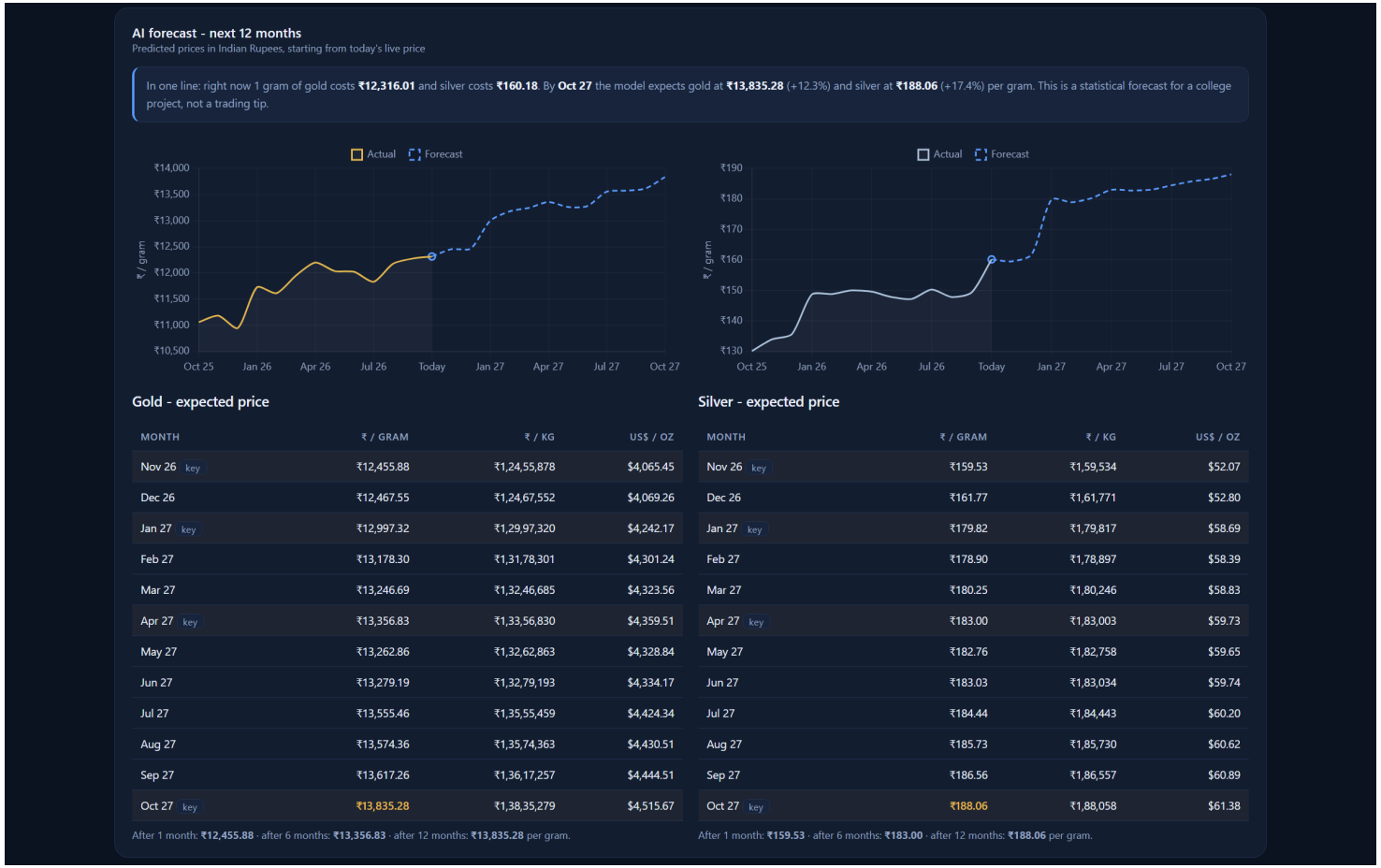


Application - Live Dashboard



Today's live gold and silver price in rupees per gram, per kg and per 10 gram, with the monthly history since 2000.

Application - 12 Month Forecast



Gold - expected price

MONTH	₹ / GRAM	₹ / KG	US$ / OZ
Nov 26 key	₹12,455.88	₹1,24,55,878	$4,065.45
Dec 26	₹12,467.55	₹1,24,67,552	$4,069.26
Jan 27 key	₹12,997.32	₹1,29,97,320	$4,242.17
Feb 27	₹13,178.30	₹1,31,78,301	$4,301.24
Mar 27	₹13,246.69	₹1,32,46,685	$4,323.56
Apr 27 key	₹13,356.83	₹1,33,56,830	$4,359.51
May 27	₹13,262.86	₹1,32,62,863	$4,328.84
Jun 27	₹13,279.19	₹1,32,79,193	$4,334.17
Jul 27	₹13,555.46	₹1,35,55,459	$4,424.34
Aug 27	₹13,574.36	₹1,35,74,363	$4,430.51
Sep 27	₹13,617.26	₹1,36,17,257	$4,444.51
Oct 27 key	₹13,835.28	₹1,38,35,279	$4,515.67

After 1 month: **₹12,455.88** · after 6 months: **₹13,356.83** · after 12 months: **₹13,835.28** per gram.

Silver - expected price

MONTH	₹ / GRAM	₹ / KG	US$ / OZ
Nov 26 key	₹159.53	₹1,59,534	$52.07
Dec 26	₹161.77	₹1,61,771	$52.80
Jan 27 key	₹179.82	₹1,79,817	$58.69
Feb 27	₹178.90	₹1,78,897	$58.39
Mar 27	₹180.25	₹1,80,246	$58.83
Apr 27 key	₹183.00	₹1,83,003	$59.73
May 27	₹182.76	₹1,82,758	$59.65
Jun 27	₹183.03	₹1,83,034	$59.74
Jul 27	₹184.44	₹1,84,443	$60.20
Aug 27	₹185.73	₹1,85,730	$60.62
Sep 27	₹186.56	₹1,86,557	$60.89
Oct 27 key	₹188.06	₹1,88,058	$61.38

After 1 month: **₹159.53** · after 6 months: **₹183.00** · after 12 months: **₹188.06** per gram.

Solid line = actual price, dashed line = model forecast, plus the month by month expected price in rupees.

Application - Model Accuracy

Model accuracy

Best model: **Random Forest**, chosen by the lowest error on 60 months it never saw.

Gold - test results

tested on the last 60 months (from Oct 2021)

MODEL	MAE US$/OZ	MAE ₹/G	RMSE US$/OZ	R2
Random Forest best	44.34	136	73.44	0.9910
Linear Regression	54.12	166	81.71	0.9889
Ridge	53.54	164	81.69	0.9889
Naive	56.93	174	86.68	0.9875

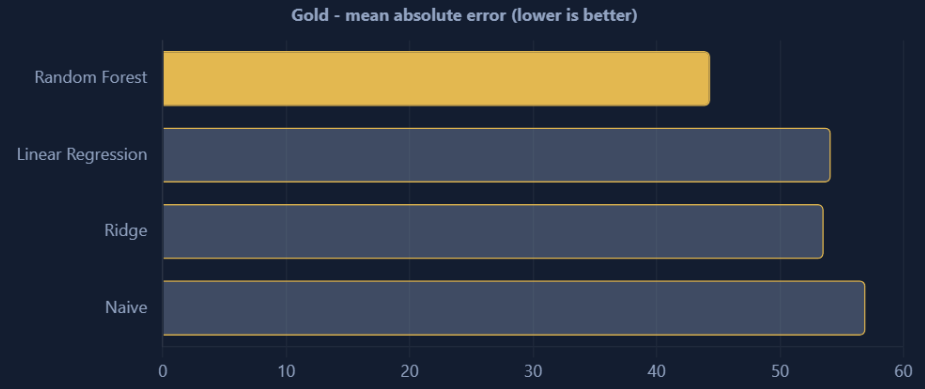
Trained on 321 monthly rows. Errors are per troy ounce in US dollars and per gram in rupees.

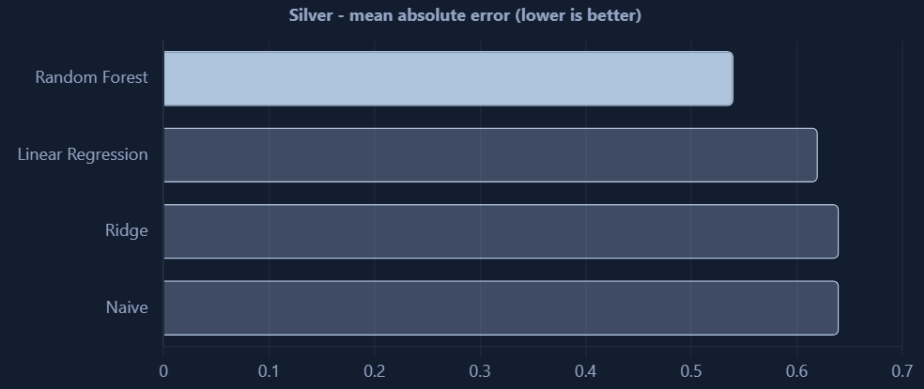
Silver - test results

tested on the last 60 months (from Oct 2021)

MODEL	MAE US$/OZ	MAE ₹/G	RMSE US$/OZ	R2
Random Forest best	0.54	1.65	0.72	0.9942
Linear Regression	0.62	1.90	0.91	0.9909
Ridge	0.64	1.96	0.94	0.9903
Naive	0.64	1.96	0.97	0.9897

Trained on 321 monthly rows. Errors are per troy ounce in US dollars and per gram in rupees.





Error of every model on 60 unseen months, compared against the naive 'next month = this month' baseline.

Conclusion

- Random Forest gave the lowest error on both metals and beat the naive baseline.
- Predicting returns instead of raw prices made the models reliable on a strongly trending series.
- Live prices, rupee conversion and the forecast are combined in one self-explanatory dashboard.
- The project demonstrates a complete Machine Learning pipeline - data, model, API, interface.

Future Scope

- Add real economic features such as inflation, the dollar index and interest rates.
- Try deep-learning time-series models like LSTM for longer horizons.
- Show a confidence range around the forecast instead of a single number.
- Extend to other metals and to jewelers' rate cards used in the local market.
- Publish the dashboard online with a scheduled daily data refresh.



Thank You

Questions & suggestions are welcome

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